PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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XXX_PocketCFO

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Table of Contents

1. Project Overview 1
2. Background & Business Objective 1
3. Product Purpose 2
4. System Functionalities 2
5. User Stories & Use Cases 3
6. Features Included (Scope Definition) 3
7. Features Not Included (Scope Control) 3
8. Assumptions & Constraints 4
9. Risks & Questions Throughout Development 4

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Project Name: PocketCFO

Version: 1.0

1. Project Overview

- **Problem Statement:** Financial management for SMEs is currently a rigid, manual, and reactive process. Business owners, especially those in the F&B or retail sectors (like a cafe owner in Gelugor), often find it difficult to visualize their cash flow health or navigate complex LHDN e-invoicing compliance without hiring expensive consultants. This leads to "intuition-based" decision-making, which often results in liquidity bottlenecks and missed tax optimization opportunities.
- **Target Domain:** Financial Decision Intelligence and Economic Empowerment for Malaysian SMEs.
- **Proposed Solution Summary:** PocketCFO removes the friction of financial planning by enabling anyone to prompt a strategic query in simple natural language (English/Malay) and receive an instant, AI-driven financial roadmap. By utilizing the Z.AI GLM to interpret unstructured expense data, the platform shifts the user's role from a Manual Data Entry Clerk (Builder) to a Strategic Decision Maker (Curator).

2. Background & Business Objective

- **Background of the Problem:** For many years, SME financial management has followed a manual, reactive, and linear pathway. A business owner typically collects physical receipts, manually records them in basic spreadsheets or ledger books, and eventually hands them over to an accountant for month-end reporting. This process is labor-intensive, prone to human error, and requires specialized knowledge of LHDN tax

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codes and e-invoicing compliance—skills that are often a barrier for entrepreneurs who want to focus on their core business operations.

- **Importance of Solving This Issue:**By accelerating the interpretation of financial data from days to seconds, PocketCFO allows business owners to test "what-if" financial scenarios (e.g., “Can I afford a new oven this month?”) and receive instant strategic feedback. Breaking the technical barrier of complex accounting software ensures that anyone, regardless of their financial background, can visualize their business’s future cash flow and stay compliant without years of specialized training.
- **Strategic Fit / Impact:**PocketCFO aligns with the shift toward AI-Native Decision Intelligence. By integrating Gemini API as the core reasoning engine, the platform transforms fragmented expense data into quantifiable economic outcomes. This facilitates a seamless transition for SMEs into the digital economy, ensuring that financial strategy—not just data entry—is at the heart of their business growth.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-enabled Strategic CFO that transforms manual financial tracking into proactive decision intelligence for SMEs.

3.2. Intended Users (Target Audience):

- **SME Owners (e.g., F&B, Retail, Service Providers):** To gain a high-level view of cash flow without needing an accounting degree.
- **Early-Stage Founders:** To manage burn rates and plan equipment or inventory investments strategically.
- **Micro-Entrepreneurs & Gig Workers:** To simplify **LHDN tax categorization** for personal and business expense separation.
- **Freelancers:** To track project-based profitability and tax deductible eligibility in real-time.
- **Non-Financial Managers:** To visualize the economic impact of departmental spending before finalizing budgets.

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4. System Functionalities

4.1. Description: The system operates as a High-Class Financial Reasoning Engine that translates natural language into strategic business decisions. Using a Prompt-to-Strategy model, the intelligence captures complex relationships between raw expenses, current cash flow health, and LHDN tax compliance requirements to generate actionable advice.

4.2. Key Functionalities:

- **Strategic "Vibe" Analysis:** Translates descriptive human language (e.g., *"I want to upgrade my kitchen but I'm worried about next month's tax"*) into high-fidelity financial roadmaps.
- **Conversational Budgeting:** Facilitates continuous modification of financial plans through iterative chat. Users can refine advice by adding context, such as changing a purchase priority or adjusting a budget limit.
- **Quantifiable Impact Projection:** The AI agent generates dozens of "What-If" variations to show how a single financial decision affects long-term runway and tax savings.
- **LHDN-Ready Export:** Enables the conversion of unstructured chat insights into structured reports and tax-ready categories compatible with Malaysian e-invoicing standards.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: : Gemini API was selected as the core engine due to its superior reasoning capabilities in specialized domains (Malaysian Finance/Tax). Its ability to handle multi-step logical deductions makes it the ideal choice for moving a user from "Data Entry" to "Strategic Curation."

4.3.2. Prompting Strategy: We utilize a Multi-Step Agentic Prompting approach. The system first extracts entities (amount, item, date), then references a "Knowledge Base" of Malaysian tax laws, and finally synthesizes a strategic recommendation. This ensures the advice is grounded in logic rather than just patterns.

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4.3.3. Context & Input Handling: PocketCFO handles unstructured data by filtering out non-financial noise. The maximum input size is capped at 1,500 tokens. If an input exceeds this limit (e.g., uploading a massive list of raw transactions), the system performs intelligent chunking, summarizing previous context to stay within the model's high-performance window while maintaining the "financial thread."

4.3.4. Fallback & Failure Behavior: If the model returns an off-topic or hallucinated response, the system triggers a Graceful Error State. An internal "Reasoning Validator" checks the output against known financial bounds (e.g., "Is the tax claim > 100%?"); if it fails, the UI requests a retry with a suggestion: "I couldn't verify that tax rule. Could you provide more details about the expense type?"

5. User Stories & Use Cases

- **User Stories:**
  - As an SME owner, I want to describe my business expenses in my own words (e.g., "Just spent RM400 on staff lunch") so that I don't have to manually figure out tax categories or spreadsheet formulas.
  - As a Founder, I want to ask about the impact of a large purchase (e.g., "Should I buy the RM5,000 espresso machine now?") so that I can make a data-driven decision based on my current cash flow.
  - As a Freelancer, I want to see a visualization of my "Cashflow Health" so that I can understand my financial runway without hiring a full-time accountant.
- **Use Cases (Main Interactions):**
  - Prompt to Strategy Flow: The user enters an unstructured financial update or query. PocketCFO parses the intent, identifies relevant LHDN tax codes, cross-references current budget data, and renders a strategic recommendation on the chat interface.

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- Impact Visualization Flow: After a decision is generated, the user triggers the "Show Impact" function. PocketCFO updates the static Dashboard Mockup, showing a visual "Before vs. After" of the business's cash flow and projected tax savings.
- Iterative Refinement: The user provides additional context (e.g., "Actually, I can pay in installments") to an existing advice thread. The AI agent recalibrates the strategy while maintaining the "financial tokens" (current balance, previous transactions, and tax year context).

6. Features Included (Scope Definition)

These are the high-impact features being built for the **PocketCFO** MVP to demonstrate technical feasibility and "Decision Intelligence":

- **Conversational Financial Workspace:** A professional chat interface where users can input unstructured expenses and receive real-time strategic advice.
- **AI-Driven Decision Logic:** Implementation of the **Z.AI GLM** to process raw text into structured LHDN tax categories and cash flow recommendations.
- **Contextual "What-If" Analysis:** The ability for the system to reason across previous transaction history to provide purchase-timing advice.
- **Static Insight Dashboard:** A "high-class" visual representation of cash flow health, highlighting quantifiable savings and tax benefits generated by the AI.
- **Natural Language to Tax-Mapping:** Support for converting common business language (English/Malay) into official tax-deductible expenditure codes.

7. Features Not Included (Scope Control)

- **Direct LHDN/Bank API Integration:** The system will not perform real-time filing or pull live bank data; these interactions are simulated for the MVP.

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- **Full Production Accounting Suite: PocketCFO** is a strategic advisor, not a full replacement for production-level double-entry bookkeeping software (e.g., SQL Ledger).
- **Multi-User / Enterprise Collaboration:** The initial version is strictly a single-user experience optimized for the individual SME owner or founder.
- **Native Mobile Application:** The prototype is built as a **Responsive Web App (Next.js)** rather than a native iOS or Android application.

8. Assumptions & Constraints

- **LLM Cost Constraint:** We estimate an average token cost of **$0.02 to $0.05 per user session** (assuming 10–15 turns of strategic dialogue).
    - **Design Decision:** To keep inference costs manageable, we implemented **Contextual Pruning**. Instead of sending the entire chat history back to the **Z.AI GLM** for every new expense, the system summarizes previous transactions into a "Financial Context Token" that remains static until a significant change in cash flow occurs.
- **Technical Constraints:** The "LHDN-Ready Export" feature is currently limited to **PDF and CSV formats**. Direct integration with cloud accounting software (like QuickBooks or Xero) or live government e-invoicing portals is not supported in this version.
- **Performance Constraints:** To maintain high-quality reasoning and avoid "token drift," there is a **daily limit of 50 strategic queries** per user. Complex multi-year financial forecasting is restricted to ensure response latency remains under **800ms**.
- **User Input:** The system is an "Advisor," not an "Automator." It cannot execute bank transfers or file taxes without explicit human approval and verification of the AI-generated data.

9. Risks & Questions Throughout Development

- **Logic Hallucination:** How do we ensure the AI doesn't invent fake tax deductions?

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- *Mitigation:* We use **Grounding**. The system prompt contains a hard-coded reference list of valid Malaysian tax categories. If the GLM suggests a category outside this list, the "Reasoning Validator" flags it.

- **Data Privacy & Security:** How is sensitive financial data protected during the inference process?
  - *Mitigation:* We utilize **PII (Personally Identifiable Information) Stripping** before data is sent to the Z.AI API, ensuring that names and account numbers are replaced with generic identifiers (e.g., [USER_A]).
- **Ambiguous Entry Interpretation:** What happens if a user provides a vague prompt like *"Spent RM50 today"* without specifying the item?
  - *Handling:* The system is designed to pause and ask a clarifying question: *"I've recorded the RM50, but to provide tax advice, I need to know if this was for office supplies, meals, or travel. Could you clarify?"*

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